

SHRIYASH PATIL

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SUMMARY

Backend Software Engineer with M.S. in Software Engineering and 2+ years designing and operating production distributed systems, event-driven architectures, and secure REST APIs. Experience shipping fault-tolerant, high-reliability systems with measurable improvements to throughput, latency, and uptime. Proficient in Python, Django, PostgreSQL, Redis, Celery, Docker, Kubernetes, and AWS; open-source contributor to NVIDIA AI-Dynamo and PyTorch.

EXPERIENCE

Mission Action | Software Engineer

Jul 2025 – Present | San Francisco, CA

- Developed a 3-tier call routing engine (direct dial, queue, voicemail) with atomic PostgreSQL SELECT FOR UPDATE SKIP LOCKED transactions, eliminating race conditions and achieving 99.9% dispatcher assignment reliability under concurrent load.
- Built a proactive dispatcher health system with 30-second Celery cleanup cycles and 2-hour lookback windows, reducing stuck-dispatcher incidents by 70% and cutting mean recovery time to under 30 seconds.
- Designed Redis-backed Celery task queue across 15+ async task types with 45-second timeout SLA; tuned 5-second cache TTL for dispatcher availability, reducing database read load by 40%.
- Implemented HTTPS, CSRF protection, secure session cookies, and token-based auth via Django; instrumented Prometheus metrics and Rollbar error tracking, cutting incident detection time by 60%.
- Containerized the stack with Docker Compose, provisioned AWS via Terraform, and implemented Pusher WebSocket real-time broadcast via Celery workers; GitHub Actions CI/CD reduced deployment errors by 75% and eliminated manual releases.

Inventech | Software Engineer

Jun 2022 – Jul 2023 | Mumbai, India

- Owned backend APIs on a React.js and Node.js e-commerce platform serving 200+ daily manufacturing users, delivering end-to-end features in two-week agile sprints.
- Designed MySQL schemas and optimized order and inventory queries; configured Docker Compose on AWS EC2 with GitHub Actions CI/CD, cutting deployment cycle from days to under 2 hours.
- Integrated Salesforce CRM with custom objects and workflow automation; introduced structured logging across backend services reducing incident detection lag by 50%.

Brightmind | Software Engineer

Feb 2025 – Jun 2025 | San Francisco, CA

- Built a Python and OpenCV OCR pipeline that extracted structured student data from 600+ scanned forms, reducing manual data entry effort by 85% and cutting processing time from hours to minutes.
- Designed a PostgreSQL ETL pipeline with idempotent upsert logic eliminating 40% of manual processing; built Flask and Chart.js real-time dashboards delivering enrollment KPIs to non-technical stakeholders.

Prime Technologies | Embedded Systems Intern

Dec 2017 – Jun 2018 | Mumbai, India

- Developed C++ firmware for PIC microcontrollers (SPI, I2C, serial) sustaining 500 sensor readings per second using OOP driver architecture; built automated calibration tooling reducing calibration time by 40%.

OPEN-SOURCE CONTRIBUTIONS

NVIDIA AI-Dynamo — Merged PR fixing missing logging import; enabled standalone module execution for 2,000+ developers [PR #1580]

PyTorch Tutorials — Fixed torch_logs tutorial: gated CUDA check and added CPU fallback for broader hardware compatibility [PR #3821]

PROJECTS

- **AI Finance Tracker** | *Next.js, Firebase, LangChain, ChromaDB, TypeScript, Chart.js, OAuth 2.0* — Full-stack AI finance app that processes transactions, stores semantic embeddings in a vector DB, and uses a LangChain RAG pipeline to answer natural-language financial questions with real-time personalized insights via Chart.js dashboards and Firebase Auth.
- **Public Health Analytics Platform** | *Python, Pandas, NumPy, Scikit-learn, FastAPI, React, PostgreSQL, Docker* — Python analytics dashboard ingesting real-time public health datasets through ETL pipelines; applied ARIMA and LSTM forecasting models to visualize infection trends and predict case growth with Grafana anomaly alerting.

EDUCATION

California State University, Fullerton | M.S. Software Engineering | GPA: 3.68/4.0

Aug 2023 – Jan 2025

Savitribai Phule Pune University | B.E. Computer Engineering | GPA: 9.24/10

Aug 2019 – Apr 2022

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C++, SQL, Go, Bash

Backend & Architecture: Django, FastAPI, Flask, Node.js, REST APIs, Event-Driven Architecture, Distributed Systems, WebSockets

Async & Messaging: Celery, Redis, Kafka, RabbitMQ, Prometheus, Sentry, Rollbar, Twilio, SendGrid, Pusher

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, CI/CD

Databases: PostgreSQL, MySQL, MongoDB, Redis, Firebase, TimescaleDB, Elasticsearch, ChromaDB, Vector DB

AI/ML & Security: LangChain, PyTorch, OpenCV, Scikit-learn, OAuth 2.0, Django Auth, CSRF/HTTPS, Token Auth, RBAC, Git